Frontier



ZeroEx

R&D application

2026

General Application

The General Application applies to everyone; all applicants should complete this section

Public section

The content in this section (answers to questions 1(a) - (e)) will be made public on the Frontier GitHub repository after we announce funding decisions in the summer and fall. Include as much detail as possible but omit sensitive and proprietary information.

Organization name

ZeroEx

Organization location (we welcome applicants from anywhere in the world)

Munich, Germany

Name(s) of primary point(s) of contact for this application

Tamara Michaelis

Brief organization description, including organization type (e.g., for-profit, NGO, research institution/university) <20 words

ZeroEx is a German project developer and technology company working to make ERW a commercially viable, scientifically rigorous CDR solution.

1. Public summary of proposed project[1] to Frontier

a. **Project title** <20 words

TRACE — ***T****racer-based* ***R****esin* ***A****ccumulator* ***C****alibration for* ***E****RW*

[1] We use “project” throughout this template, but the term is not intended to denote a single facility or experimental setup. The “project” being proposed to Frontier could include multiple facilities/locations for research activities.

b. **Project summary and field impact:** Provide an overview of your core research goal and the benefits you think it could provide to the broader CDR field. Tell us how your project meaningfully advances knowledge on at least one of the priority innovation areas identified in the RFP. What needs to happen for your project to unlock step-change advancements in cost and scale for this innovation area? Please include figures and system schematics, and be specific but concise. <1000 words

This project directly advances the path to scalable and accurate MRV laid out in the priority innovation area "Improving the MRV toolkit for open-system pathways". Our ion exchange resins, termed Self-Integrating Accumulators (SIAs), are currently close to full acceptance in established ERW methodologies (partly thanks to previous support from Frontier). This project is intended to address the principal remaining uncertainties about SIA reliability in common Global South soil types and the optimal SIA installation method for each.

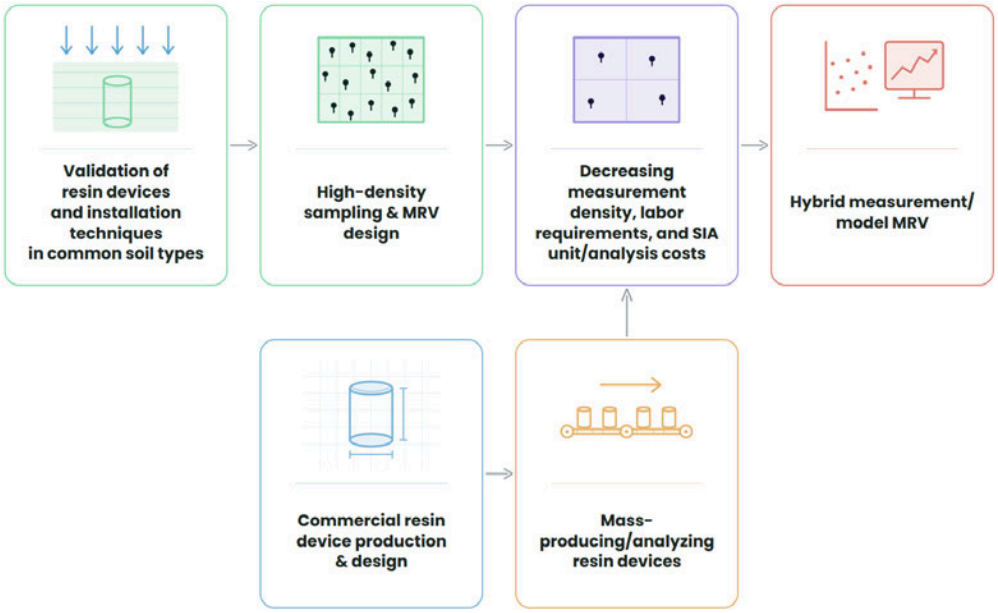


Figure 1: Development pathway of SIAs to scale as a cost-effective, globally deployed MRV tool.

The schematic above outlines the development pathway needed for SIAs to become a reliable and cost-efficient open-system MRV approach for ERW. SIA mass production is already underway in Germany, and ZeroEx is currently setting up a production and processing facility closer to projects in the Global South to decrease transportation costs and MRV overhead, with the goal of setting the SIA sale price at below $100 per unit in the future. This project aims to address the remaining uncertainties of the SIAs to generate the empirical foundation needed for significant immediate cost savings in ERW (upper track of the schematic). In the longer term, the field datasets generated here and in follow-up projects supported by this work will also be used to calibrate weathering models, contributing to the hybrid measurement/model MRV approach that would ultimately reduce physical sampling requirements even further.

In this project proposal, the first aim is to characterize whether SIA devices and their installation methods produce reliable estimates of ion flux in common ERW-relevant Global South soils. A hydraulic tracer experiment will quantify how accurately SIAs sample water flux under each combination of soil type and installation method, replicated across 5 fields in South Africa and 3 in Brazil. Past research has shown that the installation method can affect measurement results, most likely due to changes in soil structure introduced during installation, and the magnitude of this effect is expected to vary by soil type. The resulting dataset will provide field ion flux sampling efficiencies across a range of Global South soils, generating trust in the method for suppliers, buyers, and registries and supporting rapid commercial deployments at low risk.

The second aim of this project proposal is to perform a MRV intercomparison across a range of tropical soil conditions, using multiple slag-based, high-CDR-potential feedstocks. The paired soil-, lysimeter-, and SIA-based measurements generated will establish how each method performs under matched field conditions. As SIA and soil-based MRV methods already represent substantial reductions in labor and analytical requirements relative to lysimeters, they are the only cost-effective route to scaling ERW. The sampling design proposed here will identify the MRV cost per tonne of carbon removal for both methods, characterizing how commercial scaling can be realized most effectively in each soil type. In addition, testing different slag feedstocks in this design will benefit the wider sector by demonstrating that these highly reactive materials produce clear weathering signals on shorter project timelines than other feedstock types while remaining environmentally safe, improving unit economics for commercial deployments and supporting the global uptake of the ERW strategy.

Together, these outputs deliver the empirical foundation that ERW developers, registries, and buyers need to confidently adopt SIAs as a low-cost, reliable MRV method across soils in two regions relevant to ERW scaling. Realizing the step-change in ERW MRV cost requires both demonstrating SIA reliability in Global South soils and identifying which MRV method produces the most reliable and cost-efficient results in each soil type. This project delivers both, complemented by ZeroEx's parallel progress on SIA mass production and processing to ensure the cost reductions translate to commercial deployments.

a. **Project objectives:** Describe your project in more detail: what are your key research questions and proposed approach for answering them, including methods, project location, and scale where applicable? Tell us why your approach is best-in-class, and how you're differentiated from any other organization working on a similar approach. Please include figures and system schematics, and be specific but concise. 1000-1500 words

This project proposal addresses the following research questions:

- Do SIAs deliver accurate, unbiased aqueous-phase flux measurements across diverse agricultural soil types relevant to ERW scaling?

- Does soil disturbance during SIA installation introduce systematic hydraulic bias, and how does this vary with soil type/texture and installation protocol?
- What is the field-scale CDR performance of slag feedstocks in South Africa (Gauteng) and Brazil (Goiás)?

The project will answer these questions by means of high-quality data from small-scale research sites distributed across soil types in South Africa and Brazil, structured in four main work packages (Figure 2).

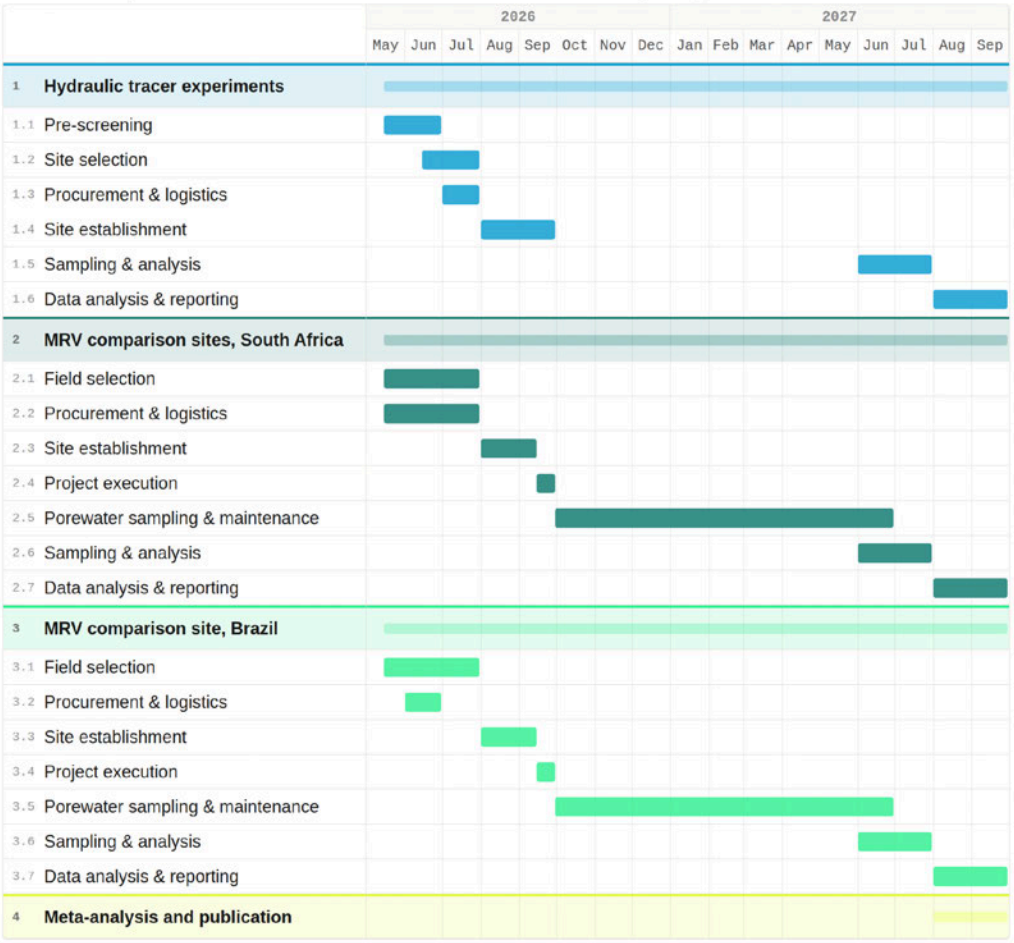


Figure 2: Gantt chart of the project timeline.

WP1: Hydraulic tracer experiments to quantify the in-field recovery efficiencies of SIAs in various soil types and for different installation methods

WP1 builds on previous research that has generated highly promising results for the use of SIAs in both laboratory and field trials. Mesocosm data has shown high adsorption efficiencies for major cations and anions and a comparison to suction lysimeters on three German fields has shown good

agreement between the two methods, unless under waterlogged conditions. This work was supported by Frontier's previous grant.

However, flow path disturbance is a fundamental challenge when installing any sensor or device in the soil profile because the installation can alter the surrounding matrix and can destroy macropore structures. Yet, this effect is rarely characterized or accounted for by developers in the ERW sector. This research component quantifies how SIA installation and associated soil disturbance can affect water flow and solute transport to and through the device. In theory, soil disturbance above or near the SIA can cause vertical porewater flux to be funneled through or diverted around the device, depending on the relative hydraulic conductivities of the surrounding soil and SIA substrate (Figure 3). This could result in either an over- or underestimation of solute flux and consequently CDR. The magnitude and direction of this effect is expected to depend strongly on soil texture, structure, and the installation technique used.

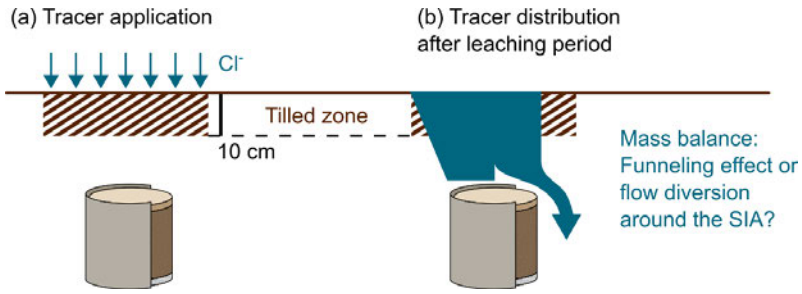


Figure 3: Illustration of the flow pattern studies with the hydraulic tracer experiment. Application of a salt tracer above the device will reveal whether water flow is funneled through or diverted around the device.

To quantify these effects, we will select 5 fields in South Africa and 3 in Brazil, install SIAs with three installation methods (12 replicates each), and apply a salt tracer above (Figure 4). After one full wet season, SIAs will be retrieved from the field and the complete soil column above and around as well as the biomass growing on top will be collected for analysis. The recovered tracer mass in the SIA, soil column, and biomass will be used to calculate the in-field capture efficiency for each installation method.

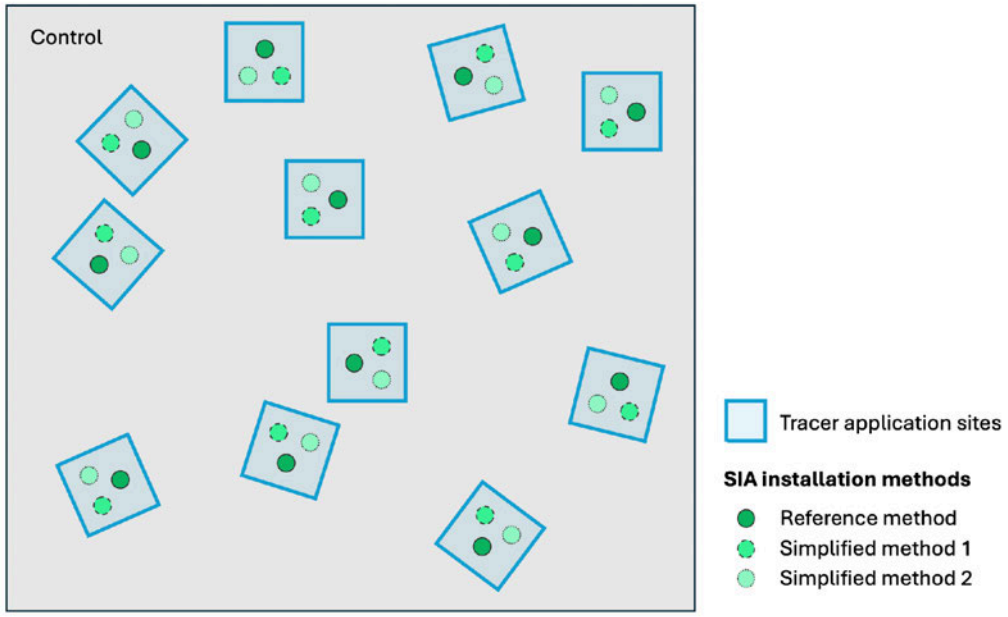


Figure 4: Conceptual design of the hydraulic tracer experiment.

WP 2: Realization of three MRV research sites with slag material in South Africa (partnering with Africa Climate Ventures)

To increase the relevance for the ERW sector, hydraulic tracer experiments will be complemented with MRV comparison sites that will yield standardized data on how accurately SIAs capture weathering signals and how they compare to both soil-based and aqueous-based CDR estimates. Highly reactive slag feedstocks create clear and fast weathering signals, which is why they are best suited for MRV research, especially when deployed in tropical or sub-tropical regions. We will utilize this advantage to expand to a region so-far unexplored for ERW: South Africa.

Early sampling and chemical analysis of the South African feedstock has identified a carbon removal potential substantially exceeding typical basalt-derived feedstocks. Critically, unlike many feedstocks with elevated CDR potential, early characterization of this material appears to carry low concentrations of all high-risk potentially toxic elements (PTEs). This combination of high CDR potential and low PTE risk is rare and makes it a strong candidate for repeated agricultural deployment.

The selected deployment region of Gauteng, South Africa, offers seasonal humid conditions that support weathering reactions, with slag stockpiles located in close proximity to agricultural land. Screening of public land use databases indicate approximately 4.2 million hectares of high-ERW-potential cultivated land within a 100 km radius of the storage site. The existing mining and processing infrastructure, alongside established logistics networks, may greatly reduce the upfront capital requirements for scaling relative to starting a project from scratch.

Within this region, we will establish three research plots designed to deliver two main outputs:

1. A direct comparison of SIA-, soil-, and lysimeter-based MRV under matched field conditions across soil and agricultural settings.
2. Time-resolved in-field slag weathering rates across soil types, identifying the most suitable conditions for future scaling.

Each trial will be set up as a small-plot block design with one high and low application rate and a control, equipped with SIAs (12 replicates per treatment) and suction lysimeters (4 replicates) for aqueous phase MRV (Figure 5). Soil-based MRV will comprise shallow (0-15 cm, 15-30 cm) and deep (1 m) sampling with a focus on cation migration through the soil. Biomass samples will be analyzed to assess plant uptake and environmental safety.

Wherever possible and within budget, we will match standards set by Cascade Climate's Coordinated Research Network (CRN) to produce datasets suitable for a wider comparison. This could for example mean using the same type of lysimeters and coordinating sampling depths and analytical protocols.

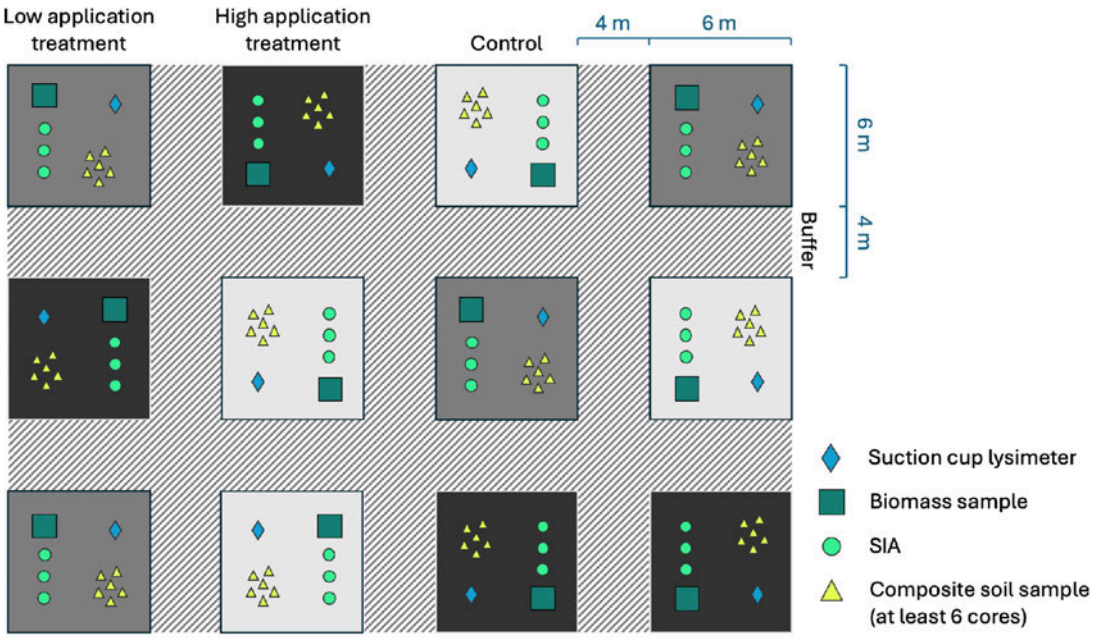


Figure 5: Conceptual design of the small-plot trial design.

WP 3: Cross-continental comparison of different slag materials with one identical MRV site in Brazil (partnering with Anglo American)

Due to the synchronized wet seasons in South Africa and Brazil, it is possible to conduct simultaneous field trials, replicating the same standardized experimental design, and compare the performance of MRV methods and slag materials across continents. This dataset will be highly

valuable for future model development as there is a strong need for comparable calibration and validation data spanning different soils, climates, and feedstocks.

For this, we will replicate the trial design described in WP2 on one field in Brazil, choosing a soil type that has proven suitable for detecting an MRV signal within one wet season in previous pilot deployments from 2024 and 2025.

ZeroEx and Anglo American have established one of the largest ERW pilot deployments globally in Goiás, Brazil, spanning nearly 4,000 ha and four farms. More than 400 SIAs are already in use as part of a creditable MRV framework using a 3-plot design, but funding for a purely scientific trial testing MRV methods and slag performance in local field conditions has so far been lacking. That said, the large amount of existing data makes it possible to narrow the scope in this region to one field, complementing previous SIA deployments, and justifies a greater focus on areas in South Africa. The existing Earthstone SIA deployments span substantial field areas and therefore can be used for defining optimal sampling densities, which is why this is not a key research goal of this proposal.

This cross-regional approach is critical to ensure that SIA-based MRV remains robust and scalable, and that SIAs can be confidently adopted as a standardized method across different regions, feedstocks, soil types and deployment practices.

WP 4: Meta-analysis and publication

A meta-analysis of all data gathered in this study, combined with other ZeroEx research sites using SIAs, will generate the body of proof showing where and how SIAs are ideally deployed as low-cost, scalable MRV method. To benefit the whole sector, we will prepare a manuscript for open access publication in a peer-reviewed scientific journal and publish our data on the Cascade Climate Data Quarry.

a. **Risks:** What are the biggest risks to your project's success and how will you mitigate those? Include scientific; technical; project execution; measurement, reporting and verification (MRV); ecosystem; financial; and any other risks. <500 words

Scientific & Technical Risks:
The primary scientific risk is that weathering signals may not be resolvable or exhibit more variability than expected across soil type, climate and installation method. This could limit the ability to compare SIA-derived flux against conventional MRV approaches, eroding confidence in method validation. This will be mitigated through rigorous data-informed site selection, replication, and standardized protocols. These measures should maximize signal to noise ratios and ensure robust and comparable datasets.

Project Execution Risks:
Real-world agricultural trials are inherently vulnerable to uncontrollable factors such as weather events, wildlife interference, changing farmer practices, or operational disruptions. To mitigate this

risk, this project involves multiple unconnected sites, so, local-scale interference at any single site does not compromise the entire dataset.

MRV Risks:
Conventional porewater measurements require assumptions such as a water balance to convert ion concentrations to fluxes, introducing uncertainties which can skew the final CDR calculation. To mitigate this risk to what is intended as the "true" measurement to which to compare the SIAs, we will use duplicate measurement devices (and have spares available if they malfunction) and run sensitivity tests on both the water balance and concentration values to quantify the inherent uncertainty.

Ecosystem Risks:
Environmental impacts from slag application, such as PTE release, dust generation during application and unintended soil and plant chemistry alteration are inherent risks of ERW projects. This is mitigated through screening of feedstock composition prior deployment, and appropriate adjustment of application rates. In addition, monitoring soil, crop tissue, and porewater will be done to detect any unexpected environmental responses early to confirm that the project remains safe within and beyond the planned operating procedures.

Financial Risks:
The voluntary carbon market for ERW is novel and remains subject to changes of certification protocols and standards. Shifts in buyer requirements or MRV protocol frameworks could affect the commercial relevance of the outputs. However, the foundational questions addressed here, including cross-continental slag comparability and soil-type suitability mapping, will remain relevant regardless of how market standards develop.

b. **Proposal to Frontier:** Please list your primary research deliverables, project timeline, and requested funding below. If you are selected for Frontier funding, this table will form the basis of contract discussions.

Primary research deliverables *(List tangible outputs of the project: e.g., validated datasets, peer-reviewed models, bench-scale prototypes, or field trial results. Should focus on resolving a specific bottleneck or uncertainty identified in Section 1.a)*	• In-field SIA flux rate efficiencies for different soil types and installation methods as last step in SIA validation • Synchronized MRV comparison data sets using standardized protocols from two regions relevant to ERW scaling • Demonstration of the efficiency of slag materials as ERW feedstock • An open access peer-reviewed publication on SIA performance in tropical and subtropical climates
Field impact *(Briefly describe how these deliverables resolve key*	The project addresses two main ERW bottlenecks: 1.) **MRV Costs**: SIAs could be a breakthrough in aqueous-phase measurements, removing the cost and operational burden of

bottlenecks for a specific CDR pathway or broader CDR field, e.g. efficiency gains, scaling unlocks, or reducing measurement uncertainty.)	conventional aqueous phase sampling. 2.) **Feedstock reactivity and resolvability of weathering signals**: Slag materials can have high CDR potentials and are typically more reactive than basalt feedstocks, ensuring good resolvability and short project timelines
Project timeline & key milestones *(At what point should Frontier consider the research project complete? Include major progress markers. Frontier generally prefers projects with clear readouts within ~18 months.)*	Key milestones: **Q2 2026**: Site selection completed in South Africa and Brazil. **Q3 2026**: SIA and lysimeter installation completed. **Q3 2026**: Feedstock spread on MRV comparison sites, pre- and post-spreading soil samples collected. **Q2/Q3 2027**: Post-weathering soil samples collected, biomass samples collected, and SIAs retrieved. **Q3 2027**: SIA installation protocol recommendations finalized. Slag CDR performance comparison completed. **Q4 2027**: Manuscript for peer-reviewed publication prepared.
Total grant request (US$) *(The total funding required to execute the research scope described above. Frontier R&D grants typically range from US$250k–$750k.)*	$ 745,329